MRI Install Form

1850 Town and Country, Norco, CA 92860 USA
Main: 951.734.3737, Fax: 951-735-3373

OEM ID		Van ID	
Site Name		City, State	
Site Contact		Phone Number	
SMR Rep		Inspection Date	

Pre-Site inspection

Engineer ______ Date ______

Site Name ______

Street Adress ______

Shipping Adress ______ Loading Dock ______

City ______ State ______ Zip Code ______

Walk Site to identify any potential problems

- [ ] Identify Magnet Enterance path, any problems, steel beams, window cracks, height, width, corners, and or soft ground
  - [ ] Verify with Rigger

  Magnet Opening Width; min 7ft ______ Magnet Opening Height; min 7ft ______

  Type Serial # Weight (T) Size L X W X H H includes 6in Skates

  Magnet Description ______

- [ ] Identify Equipment Enterance path, doorways, raised floor, loading dock, pallet jack access, staging area
  - [ ] Verify with Rigger

  Door Height; min 84in. ______ Door Width; min 40in. ______

- [ ] Identify cryogen vent connection, location, exit and size — Cryogen Vent Diameter ______
- [ ] Identify Magnet mounting, anchor bolts or Vibro Pad — Anchor bolt Width ______ Anchor bolt Lenght ______
- [ ] Identify Pen Pannel opening location size and mounting — Pen Pannel Width ______ Pen Pannel Hight ______
- [ ] Identify main disconnect power and location — Voltage ______ Current ______
- [ ] Identify DC Lighting location and power - Filters through screen room — Voltage ______ Current ______
- [ ] Identify water chiller location, power and water — Length ______ Width ______
- [ ] Identify MRU location (need 120V power or outlet and place to run a cable to pen pannel
- [ ] Identify Console cable run and length - minimum 2in pipe for power and 3in pipe for data cables (no sharp sweeps)

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Verify site is ready for magnet delivery

- [ ] Verify Magnet Enterance path, any problems, steel beams, window cracks, height, width, corners, and or soft ground
  - [ ] Verify with Rigger — Magnet Opening Width; min 7ft [ ] Magnet Opening Height; min 7ft [ ]

Type Serial # Weight (T) Size L X W X H H = W/6in Skates

Magnet Description [ ]

- [ ] Verify Equipment Enterance path, doorways, raised floor, loading dock, pallet jack access, staging area
  - [ ] Verify with Rigger — Door Height; min 84in. [ ] Door Width; min 40in. [ ]
- [ ] Verify cryogen vent connection, location, exit and size — Crygen Vent Diameter [ ]
- [ ] Verify Pen Pannel opening location size and mounting — Pen Pannel Width [ ] Pen Pannel Hight [ ]
- [ ] Verify main disconnect power and location — Voltage [ ] Current [ ]
- [ ] Verify DC Lighting location and power - Filters through screen room — Voltage [ ] Current [ ]
- [ ] Verify water chiller location, power and water w/glycol — Location Length [ ] Location Width [ ]
- [ ] Verify MRU location (need 120V power or outlet and place to run a cable to pen pannel
- [ ] Verify Console cable run and length - minimum 2in pipe for power and 3in pipe for data cables (no sharp sweeps)

Moving Magnet and Equiptment

- [ ] Make sure You have fork lift(s) pallet jack and or crane to move magnet and equiptment
- [ ] Before Riggers move magnet make sure to put the shim lead down — Magnet Pressure: [ ]
- [ ] Supervise Equipment Moving and Magnet Rigging into the building
- [ ] Verify Magnet Secure and S/C Lead Extension is Dis-Engaged.
- [ ] Verify all pallets and equipment are moved into the site and in a safe clean location or covered
- [ ] Sign off Rigger [ ] Sign off Shipper
- [ ] Place compressor in equipment room, Install coldhead Lines, water from chiller and Power from Main Disconnect
- [ ] Check main input power voltage with DVM (leg to leg 480V (+or- 10V) and each leg to ground 240 (+or- 10V))
- [ ] As long as main input power is good and water chiller is running you can turn on power to the compressor
- [ ] Install Magnet Monitor or helium meter — Helium Level: [ ] Magnet Pressure: [ ]

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System Physical Installation

- ☐ Move Cabinets into place in the equipment room, check for best placement and service space
  - ☐ PDU closest to main disconnect, If possible ☐ RF and Gradient cabinets need room to pull out componets
- ☐ Install pen panel, make sure to fix helium gas lines and coldhead power cables through the pen panel
- ☐ Start installing cables by separating them out into magnet room (blue), console (Green or brown) Equipment (the rest)
- ☐ Magnet (blue), console (Green and brown), Equipment (orange, blue/green, red, and orange/yellow) and pen pannel (yellow
- ☐ Make sure to leave a 2ft service loop for the RF cabinet SSM module connections
- ☐ Use cable map to check off all connections and make sure you have all cables
- ☐ Layout console to equipment room cables to make sure of length before running (green or brown is console side)
- ☐ Make sure to make a list of any cables that are needed or missing parts
- ☐ Install MRU on wall, hook up to 120V power or outlet, run MRU cable
- ☐ Make sure you have the Last Best Know shim currents and center frequency of the magnet Magnet Serial # ______
- ☐ Make a list of anything that is missing or needs to be completed as you are working and double check as you finish
- ☐ Re-Check Helium Level and pressure of magnet before leaving site Helium Level: ______ Magnet Pressure: ______

Power Up

- ☐ Verify all power and ground connections to each cabinet ☐ Verify all cable connections to all equipment and magnet
- ☐ Make sure all breakers are turned off to equipment and the main
- ☐ Check main input power voltage with DVM (leg to leg 480V (+or- 10V) and each leg to ground 240 (+or- 10V))
- ☐ As long as main input power is good you can turn on the main breaker, re-check power from cabinet
- ☐ Turn on gradient cabinet fan first to make sure door sucks closed, if door blows open turn power off and swap 2 input phases
- ☐ If E-stop doesn't reset w/bypass then the phase may need to be changed back and phase swapped on gradients (ACGD cabinet)
- ☐ Make sure that system comes all the way up, check table movement, height and home sensor flag
- ☐ Run head and body noise scans, check standard deviation and means for receive signal using an ROI under the image viewer

Head noise scan	Standard Deviation		Mean value	
Body noise scan	Standard Deviation		Mean value	

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Calibrations

- ☐ Before ramping magnet make sure to de-ice and that shim lead connection is very clean

Helium Level above 85% ☐

- ☐ Make sure you have the last best known shim currents and center frequency
- ☐ Follow the ramp procedure for the type of magnet you are working on in the magnet manual
- ☐ Ramp up magnet to the last best known center frequency or closest to the Tesla of the magnet, with AX shim currents
- ☐ After Ax currents are in and the ramp probs are out put in T1 and T2 shim Currents
- ☐ Run first image test scan, same as iso center, check for GE logo orientation (should be in the top right corner)
- ☐ Supercon Shim magnet until the Inhomogeneity is under 100hz and coefficients are under 100 as well
- ☐ Run L-coil, DC offsets, Iso center and gradient cals (if twin speed run zoom mode as well)
- ☐ Run grafidy in all three planes X,Y and Z until all are black or green (if twin speed run zoom mode as well)
- ☐ Re-Run L-coil, DC offsets, then re supercon shim until inhomogeneity under 50hz, check for coefficients to pass specs
- ☐ Re-Run L-coil, DC offsets, Iso center and gradient cals (if twin speed run zoom mode as well)
- ☐ Re-Run grafidy in all three planes X,Y and Z until all are green (if twin speed run zoom mode as well)
- ☐ Run gradient shim until your currents don't change any more
- ☐ Run Echo Planner Imagining scans for head and body (EPT)
- ☐ Check multicoil scans using CTL coil, scan CLTTOP and CTLBOT (if 8ch or 16ch system make sure to check all receivers)
- ☐ Setup network information including PACS, worklists and camera

System Final Installation

- ☐ Make a list of all coils on site and check against the quote of coils sold
- ☐ Make sure to test all coils for signal on each channel
- ☐ Clean up and put every thing away the best that you can
- ☐ Test network Connections and camera to make sure they are working
- ☐ Make a list of anything that is missing or needs follow up
- ☐ Run one last multi-coil, body and head scan before leaving the system

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SiTE LAYOUT

NOTES

NEEDS

Please fill out layout, notes and needs sections on last page as information is recived